

RISCBench: Benchmarking RISC-V Orchestration Efficiency in FPGA and FPGA-Like Computing Engines

An Industry Evaluation of Control-Plane Bottlenecks and Sustained Throughput Metrics

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ABSTRACT

Heterogeneous systems increasingly rely on RISC-V cores as orchestration engines, managing data movement, synchronization, and scheduling across accelerators and reconfigurable fabrics. Conventional performance metrics—FLOPs, TOPS/W, or energy per operation—do not capture orchestration efficiency, even though it often dictates sustained throughput. This gap is particularly relevant at the current industry inflection point: recent announcements of CPU–GPU co-integration, such as Intel–NVIDIA collaborations and AMD’s follow-on FPGA–GPU initiative, highlight that orchestration overhead will determine whether co-packaged fabrics achieve their promised performance.

We present RISCBench, a kernel benchmark suite and open methodology for quantifying orchestration efficiency. The benchmark introduces the Sustained Instantaneous Throughput (SIT) metric, which measures how long a system maintains near-aggregate throughput under SRAM residency conditions. RISCBench has been applied to two representative platforms: an Altera Agilex FPGA with Nios-V soft RISC-V cores, used in the prototyping stage to validate portability, and the Tenstorrent Wormhole accelerator with 320 hard RISC-V control cores, used for detailed evaluation. Expanded FPGA measurements will be included in the camera-ready version. Early results highlight synchronization and memory-residency tradeoffs between soft and hard implementations. We share lessons learned and propose SIT as a practical descriptor that may guide future FPGA and accelerator integration.

CCS CONCEPTS

- Hardware → Reconfigurable logic and FPGAs
- Computer systems organization → Heterogeneous (hybrid) systems
- Hardware → Performance evaluation

KEYWORDS

Orchestration, Efficiency, RISC-V, Benchmarking, Heterogeneous Integration

1 Introduction

Heterogeneous computing has become the dominant paradigm across datacenter, edge, and embedded systems. Rather than relying on monolithic CPUs or GPUs, designers now assemble systems from specialized fabrics: FPGAs for reconfigurable dataflow [1], ASIC accelerators for dense linear algebra [2], and lightweight cores for orchestration [3][4]. Within this landscape, RISC-V has emerged as the control-plane substrate, managing synchronization, data movement, and task scheduling across heterogeneous subsystems. Its influence is increasingly decisive: even with abundant compute resources, sustained throughput collapses if orchestration falters.

Conventional performance metrics, however, provide little visibility into this dimension. FLOPs, TOPS/W, and energy per operation capture raw compute and efficiency but overlook orchestration overhead [5][6][7]. For many real workloads, especially in FPGA-accelerated systems, orchestration efficiency—not raw compute density—determines achievable sustained performance.

This limitation is magnified at the present industry inflection point. Recent announcements of CPU–GPU co-integration, including the Intel–NVIDIA collaboration and AMD’s follow-on initiative to integrate FPGA and GPU fabrics, mark a trajectory toward tighter packaging of heterogeneous subsystems [8][9]. While such coupling reduces physical latency, it does not resolve the underlying bottleneck: the effectiveness of control-plane cores in managing memory residency, synchronization, and workload transitions across fabrics. Without a practical yardstick for orchestration efficiency, there is a risk that co-integrated devices will deliver impressive peak specifications while falling short in sustained real-world throughput.

This paper introduces RISCBench, a kernel benchmark suite and open methodology that addresses this gap. RISCBench formalizes orchestration as a measurable dimension of performance through

the Sustained Instantaneous Throughput (SIT) metric, which quantifies how long systems can maintain near-aggregate throughput under SRAM residency conditions. The benchmark has been validated through prototyping on an Altera Agilex FPGA with Nios-V soft RISC-V cores and evaluated in detail on the Tenstorrent Wormhole accelerator, an FPGA-like computing engine embedding 320 hard RISC-V cores [10][11]. These results expose orchestration inefficiencies invisible to FLOP- or energy-centric benchmarks, offering lessons for FPGA toolflows, runtime design, and heterogeneous system architecture.

This paper is aimed at the Industry Track, and it emphasizes insights and practical relevance rather than academic novelty.

2 Related Work

Benchmarking has long been central to the evaluation of new computing architectures. FLOP- and energy-centric benchmarks such as LINPACK, MLPerf, and Green500 have established widely accepted descriptors for peak throughput, energy efficiency, and scaling behavior [12][13][14]. These metrics have shaped architectural design choices and industry adoption, but they primarily reflect raw compute density rather than orchestration efficiency.

Within the FPGA community, prior benchmark suites have focused on application kernels (e.g., linear algebra, image processing, streaming workloads) or on the evaluation of CAD toolflows [15][16]. While these studies provide insights into datapath efficiency and compilation flows, they do not explicitly measure synchronization, memory residency, or control-plane overheads. Existing FPGA performance evaluations therefore remain incomplete when applied to heterogeneous systems where orchestration plays a decisive role.

The emergence of RISC-V as a ubiquitous control substrate has further highlighted the need for orchestration-aware benchmarking. FPGA soft RISC-V cores are increasingly used for scheduling and data movement, while accelerator-class platforms such as Tenstorrent Wormhole embed clusters of hard RISC-V cores for large-scale orchestration [17]. Despite this growing role, there is no established benchmark methodology that evaluates orchestration efficiency across both soft and hard RISC-V environments.

RISCBench addresses this gap by introducing a framework for evaluating orchestration efficiency through lightweight, portable kernels and a new sustained throughput descriptor. Unlike existing suites, it explicitly targets the dimension of sustained orchestration efficiency in heterogeneous FPGA and accelerator platforms.

3 RISCBench Design Methodology

RISCBench was developed to isolate orchestration as a measurable performance dimension while remaining lightweight, portable, and reproducible across FPGA and accelerator platforms. The design philosophy emphasizes clarity of orchestration effects over application-specific optimizations.

Unlike traditional FPGA benchmarks, RISCBench emphasizes control-plane behavior. The suite is not intended to measure raw datapath performance, but instead to expose orchestration overheads that directly impact sustained throughput. This distinction makes RISCBench complementary to existing FLOP- and energy-centric benchmarks.

3.1 Design Goals

The benchmark suite is guided by the following principles:

- **Simplicity** — kernels are compact, isolating synchronization and memory effects without application-level confounders. These kernels include matrix-multiplication of various sizes and memory operations of different transfer sizes.
- **Reproducibility** — experiments can be replicated across FPGA soft RISC-V cores and accelerator-class hard RISC-V clusters. Developers can share and publish results to the RISCBench leaderboard.
- **Portability** — implementations rely on minimal dependencies, enabling deployment across vendor toolflows and heterogeneous platforms. To facilitate this, we leverage containers and microservices.
- **Open Source** — the methodology is accompanied by open-source code to enable transparency, sustainability, and community contributions. The code will be available on GitHub with new workloads and specs published regularly at www.riscbench.com.

3.2 Kernel Categories

RISCBench kernels target fundamental orchestration behaviors rather than complex applications. The current suite includes:

1. **Streaming memory bandwidth** — measures sustained DRAM ↔ SRAM throughput under continuous access
2. **Streaming communication latency** — characterizes interconnect latency and jitter between cores or tiles
3. **Sustained throughput** — measures real throughput with SRAM-resident data, contrasting with theoretical peak specifications
4. **Sustained Instantaneous Throughput (SIT)** — evaluates how long a system maintains near-aggregate throughput under SRAM residency conditions, as depicted in Fig 1.

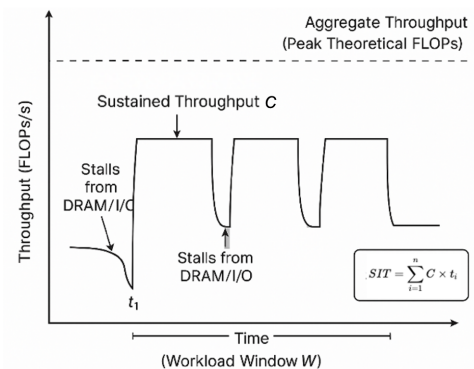


Fig 1: Illustration of Sustained Instantaneous Throughput (SIT) metric in RISCBench

3.3 How to Measure SIT

Sustained Instantaneous Throughput (SIT) is measured using RISCBench kernels, such as matrix multiplication operations of a given size and bit precision (e.g., two 64x64 16-bit matrix), executed over a workload window W .

During this window, compute units operate at or near their vendor-defined peak throughput (Aggregate Throughput) whenever data is fully resident in fast on-chip SRAM or BRAM. In these intervals, throughput is ceilinged by the theoretical peak FLOPs across all available cores.

At discrete intervals ($t_1, t_2, t_3 \dots$), the kernel achieves bursts of *Sustained Throughput*, defined as the realized throughput with SRAM-resident data. Each burst corresponds to an interval of high-efficiency computation before stalling occurs due to *Streaming Memory Bandwidth* (DRAM $\leftrightarrow$ SRAM transfers) or *Streaming Communication Latency* (chip $\leftrightarrow$ chip messaging). Throughput declines until the next SRAM residency period is reached.

Formally, SIT is expressed as:

$$SIT = \sum_{i=1}^n C \times t_i$$

where C is the compute throughput (FLOPs/s), and t_i are the instantaneous intervals of SRAM residency during execution. The formulation parallels the energy relationship $\text{Energy} = \text{Power} \times \text{Time}$, but here sustained throughput is accumulated only across SRAM-resident windows.

4. Evaluation Platforms

RISCBench was developed to operate across heterogeneous orchestration substrates, beginning with FPGA-based soft RISC-V cores and extending to accelerator-class platforms with embedded hard RISC-V clusters, as shown in Table 1. This section outlines the two representative platforms considered in this work.

4.1 Altera Agilex FPGA (Prototype Platform)

The initial prototyping of RISCBench was performed on an Altera Agilex 5 FPGA using the Nios-V soft RISC-V core. This stage was used to validate the portability of the benchmark suite within FPGA contexts, ensuring that the kernels could be integrated through vendor toolflows and executed on constrained softcore environments (Fig 2-Top). While comprehensive measurement results are not included in this submission, the FPGA prototype confirmed that RISCBench can be deployed within reconfigurable fabrics. Expanded FPGA evaluation data will be provided in the camera-ready version of this paper.

4.2 Tenstorrent Wormhole Accelerator (Evaluation Platform)

Detailed evaluation was conducted on the Tenstorrent Wormhole accelerator, a representative example of an FPGA-like computing engine. Wormhole integrates 320 RISC-V control cores distributed across 64 accelerator tiles, using them to orchestrate data

Table 1: Platform characteristics of our evaluation testbeds.

Platform	Core Type	Core Count	Memory Hierarchy	Peak Bandwidth (datasheet)	Peak Throughput (datasheet)
Altera Agilex 5 (Nios-V softcore)	Soft RISC-V core	1 (configurable)	On-chip BRAM + external DDR4	~32.4 GB/s (DDR5 single channel)	N/A (control-core limited)
Tenstorrent Wormhole	Hard RISC-V cores embedded in ASIC tiles	320 (5 per tile $\times$ 64 tiles)	Per-tile SRAM + shared DRAM	288 GB/s (GDDR6, bidirectional); 192 GB/s per core (NoC B/W)	466 TFLOPS (int8, theoretical)

movement, synchronization, and workload scheduling. As shown in Fig 2-Bottom), this architecture provides an opportunity to assess orchestration efficiency at scale, with workloads that highlight the interaction between synchronization overhead, memory residency, and sustained throughput. The availability of performance counters and profiling support on Wormhole enables more detailed analysis than was feasible during the FPGA prototyping stage.

4.3 Towards Cross-Substrate Benchmarking for Emerging Co-Integrated Architectures

The progression from FPGA prototyping to accelerator-scale evaluation demonstrates the portability of RISCBench across orchestration substrates. Soft RISC-V cores on FPGA validate feasibility, while hard RISC-V cores in Wormhole provide quantitative insight at scale. Together, these platforms illustrate the

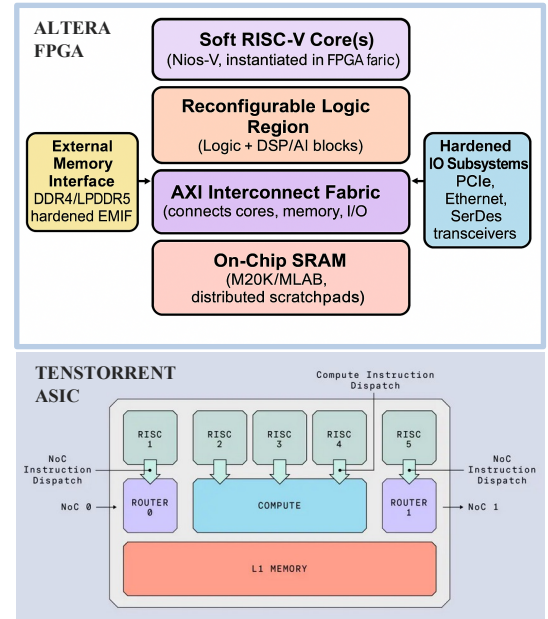


Fig 2. Architectural overview of RISCBench testbeds. *Top*: Altera Agilex 5 FPGA with a single instantiated RISC-V softcore. *Bottom*: A single Tenstorrent Wormhole tile with 5 hard RISC-V cores. With 64 total tiles, Wormhole has a combined 320 cores per device, all dedicated to orchestration – not compute.

range of environments where orchestration efficiency must be measured and establish a basis for applying RISC Bench to future co-integrated architectures (i.e., FPGA–GPU, GPU–CPU, FPGA–Quantum).

5. Experimental Results

The evaluation of RISC Bench was conducted in two stages. The benchmark suite was first deployed on an Altera Agilex 5 FPGA with a Nios-V soft RISC-V core to validate portability and integration within reconfigurable contexts. This prototyping stage confirmed that the kernels could be implemented within vendor toolflows and executed on constrained softcore environments, but quantitative measurements remain limited at this stage. Expanded FPGA results are planned for inclusion in the camera-ready version.

Detailed results are reported for the Tenstorrent Wormhole accelerator, which provides a representative example of an FPGA-like computing engine with embedded RISC-V orchestration cores. Wormhole integrates 320 RISC-V control cores across 64 accelerator tiles, enabling a detailed study of synchronization behavior, memory residency, and sustained throughput at scale.

5.1 Kernel Measurements

For RISC Bench on the Tenstorrent Wormhole testbed, execution is profiled at the granularity of cycles, zones, tiles, and buckets as shown in Fig 3. Each workload is decomposed into compute zones (loops of SRAM read–compute–SRAM write), with inputs tiled into 32×32 blocks to match vector engine requirements.

The complete run is divided into time buckets (≈ 97 cycles each in this instance), enabling measurement of instantaneous FLOPs throughput across execution. From these traces, we extract sustained throughput in the plateau region and compute SIT (Sustained Instantaneous Throughput) as the accumulated throughput while data remains SRAM-resident, providing a fine-grained view of orchestration efficiency on Wormhole’s RISC-V–driven architecture.

5.1.1 Streaming memory bandwidth. Measurements capture DRAM $\leftrightarrow$ SRAM throughput under sustained access. Results in Fig 4 indicate that orchestration efficiency significantly influences how close sustained bandwidth approaches vendor-specified peaks.

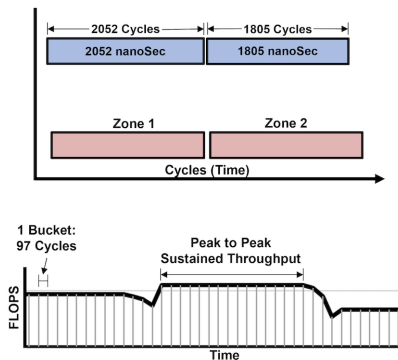


Fig 3. Kernel Measurement Granularity: Cycles, Zones, and Buckets

5.1.2 Streaming communication latency. Characterization of inter-tile synchronization latency remains under development due to current vendor limitations in available counters, and will be included in extended evaluations.

5.1.3 Sustained throughput. Sustained performance was measured across multiple execution windows, showing an early peak followed by a decline. This decline is attributed to synchronization overhead accumulating as orchestration cores saturate memory bandwidth. The effect is consistent with double-buffering strategies initially masking overhead before degradation sets in.

5.1.4 SIT metric. The Sustained Instantaneous Throughput metric provides a cross-platform yardstick, quantifying how long throughput remains near peak levels under SRAM residency conditions. SIT highlights architectural differences that are not visible through FLOP- or energy-centric descriptors.

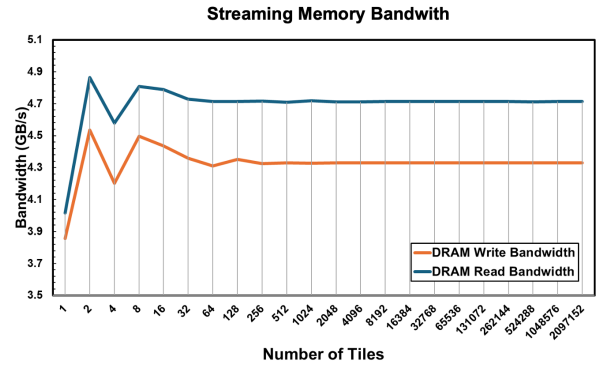


Fig 4. Streaming memory bandwidth from 2 KB to 4 GB transfers (2 KB tiles). Reads consistently exceed writes, with a brief anomaly at 2–4 tiles likely due to pipeline or controller effects, before stabilizing near 4.7 GB/s (read) and 4.3 GB/s (write), well below the theoretical limits (see Table 1).

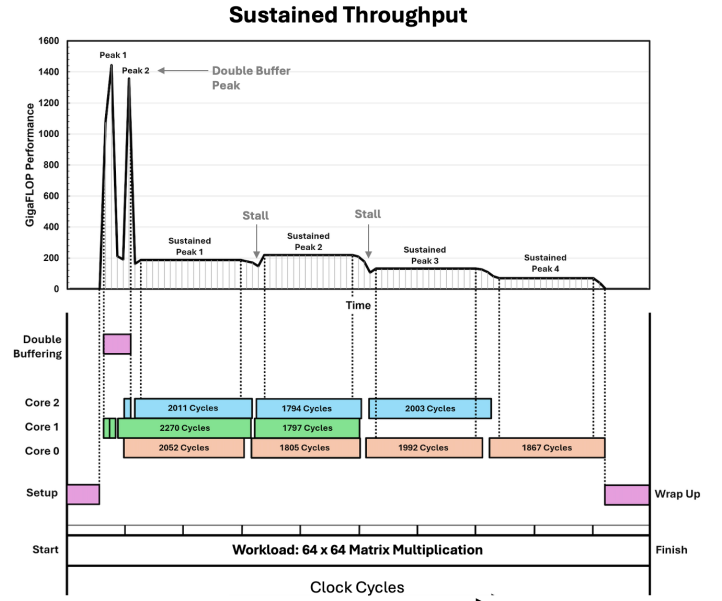


Fig 5. Sustained Instantaneous Throughput (SIT) profile in 64×64 matrix multiplication

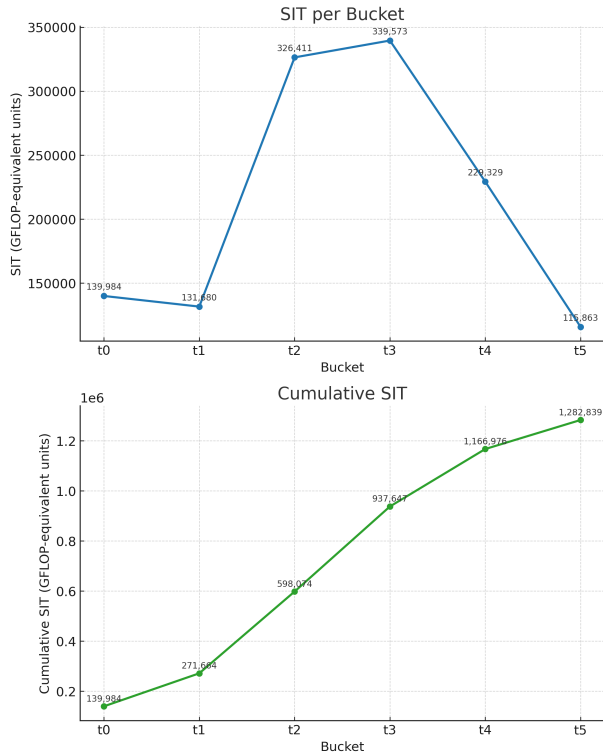


Fig 6. Per-Bucket and Cumulative Sustained Instantaneous Throughput (SIT), measured in *SIT-OPS* (or SIT-aware GFLOPS)

Figure 6 shows SIT both per bucket and cumulatively. Per-bucket SIT climbs to ~339k SIT-OPS (SIT-aware GFLOP equivalent) at t3 before declining at t4 and t5. This decline reflects synchronization overheads and residency shifts that increasingly limit sustained throughput. The cumulative SIT curve continues to rise, but with a visibly reduced slope after t3, confirming that later buckets contribute less incremental efficiency. In combination, the per-bucket and cumulative views highlight that SIT not only quantifies total sustained throughput (~1.28M units), but also pinpoints where orchestration bottlenecks begin to dominate execution.

5.2 Platform Comparisons

5.2.1 Agilex FPGA (prototype stage). The Agilex FPGA with Nios-V soft RISC-V cores confirmed portability and feasibility of RISCBench kernels in reconfigurable fabrics. While comprehensive performance measurements are not yet available due to platform limitations, early experiments suggest that orchestration bottlenecks—particularly synchronization overhead—are likely to mirror trends seen on Tenstorrent platform. Expanded results will be provided in the camera-ready version.

5.2.2 Tenstorrent Wormhole. Wormhole demonstrated orchestration scalability with hard RISC-V clusters but revealed bandwidth-driven ceilings. The initial throughput peak arose from double buffering in DST registers—data movement and compute operated in parallel at startup, then desynchronized, introducing

stalls and reducing sustained performance. Matrix multiplication showed a similar peak-drop pattern, with further loss from DRAM pressure and profiler overhead. These results underscore that synchronization and memory residency, not peak compute density, define sustained throughput on Wormhole.

5.3 Key Takeaways

RISCBench highlights orchestration inefficiencies that traditional FLOP- or energy-based benchmarks cannot capture. On FPGA soft RISC-V cores, feasibility and portability are validated, while on accelerator-class hard RISC-V clusters, sustained throughput is shaped by double-buffering stalls, DRAM pressure, and synchronization overhead. Across both substrates, SIT emerges as a platform-independent descriptor of orchestration efficiency, offering a unifying measure that can extend naturally to future co-integrated architectures.

6. Industry Lessons Learned

The evaluation of RISCBench across FPGA and FPGA-like platforms highlights several lessons that are directly relevant to industry adoption of heterogeneous computing systems.

6.1 Orchestration as a First-Class Performance Constraint

Both platforms demonstrate that orchestration efficiency is as critical as raw compute density. FLOPs and energy descriptors provide useful bounds, but sustained performance is limited by how effectively control-plane cores manage synchronization and data movement. Without orchestration-aware metrics, peak values provide little predictive value for deployed workloads.

6.2 Distinct Tradeoffs Between Soft and Hard RISC-V Orchestration

Prototype measurements on the Agilex FPGA confirm that soft RISC-V cores provide flexibility and rapid prototyping of orchestration strategies, but scaling is constrained by synchronization costs and memory access overheads. In contrast, the Tenstorrent Wormhole shows that hard RISC-V orchestration cores scale control more effectively across tiles, but throughput is capped by sustained bandwidth and memory residency. Both implementations reveal bottlenecks that are not visible through conventional benchmarking approaches.

6.3 Memory Residency Amplifies Orchestration Inefficiency

Results demonstrate that throughput can decline significantly after an initial peak, as synchronization and memory access patterns accumulate. This behavior underscores the importance of SRAM residency in sustaining high performance. SIT captures this relationship directly by quantifying the time window during which throughput remains near peak levels.

6.4 Implications for FPGA Toolflows and Runtime Systems

The absence of orchestration-aware descriptors in current FPGA toolflows and runtime systems creates risk in evaluating heterogeneous platforms. As industry moves toward co-integrated CPU-GPU and FPGA-GPU packages, failing to account for

orchestration bottlenecks may lead to designs that meet theoretical specifications but fail under sustained workloads. Metrics such as SIT can provide actionable visibility for both hardware designers and toolchain developers.

6.5 Toward a Platform-Independent Descriptor

SIT offers a common metric that unifies evaluation across FPGA prototypes and accelerator-class platforms. By quantifying orchestration efficiency independently of architecture or vendor, SIT provides a basis for comparing heterogeneous designs and for guiding integration of future co-packaged systems.

7. Roadmap and Broader Relevance

The evaluations presented in this paper position RISC Bench not only as a benchmarking methodology for current FPGA and accelerator platforms, but also as a foundation for guiding the design of future co-integrated architectures.

7.1 Hybrid FPGA+Accelerator Testbed

The next stage of RISC Bench development will involve deployment on a hybrid testbed coupling an FPGA and an accelerator via high-speed interconnect. Such a platform enables orchestration tradeoffs to be studied across substrates in real time, stress-testing synchronization mechanisms and memory residency across distinct fabrics. This hybrid configuration serves as a practical proxy for evaluating the performance characteristics of forthcoming co-packaged devices.

7.2 Proxy for Industry Trends

Recent announcements highlight the trajectory of heterogeneous integration. The Intel–NVIDIA collaboration on CPU–GPU co-integration and AMD’s initiative to integrate FPGA and GPU fabrics indicate that orchestration will increasingly determine the realized performance of future packages. RISC Bench provides an open framework for quantifying orchestration efficiency in advance of silicon availability, allowing potential bottlenecks to be identified and mitigated during design.

7.3 Applicability Beyond Current Platforms

The principles embodied in SIT extend beyond today’s FPGA and accelerator platforms. The metric is equally applicable to CPU–GPU packages, FPGA–GPU co-integrated devices, FPGA–ASIC heterogeneous systems, and FPGA–Quantum architectures [18]. As integration density increases, orchestration bottlenecks will persist, and metrics such as SIT will provide a consistent means of evaluation across diverse architectures.

7.4 Community Adoption

RISC Bench is intended to be an open, community-driven framework. Results are hosted at www.riscbench.com/results, where industry and academic contributors can publish measurements across platforms and workloads. This open leaderboard model encourages transparency, enables comparative analysis, and accelerates the refinement of orchestration-aware

metrics. By grounding the discussion in reproducible results, the community can collectively shape both the benchmark and the architectures it evaluates.

8. Conclusion

This paper introduced RISC Bench, a kernel benchmark suite and open methodology for quantifying orchestration efficiency in heterogeneous systems. The framework defines the Sustained Instantaneous Throughput (SIT) metric, which measures how long a platform sustains near-aggregate throughput under SRAM residency conditions.

RISC Bench was validated in two stages: prototyping on an Altera Agilex FPGA with Nios-V soft RISC-V cores, and detailed evaluation on the Tenstorrent Wormhole accelerator with 320 hard RISC-V control cores. Results reveal orchestration inefficiencies that are not visible through FLOP- or energy-centric metrics, underscoring the need for descriptors that capture sustained, real-world performance.

The findings provide practical insights for FPGA toolflows, runtime systems, and heterogeneous platform design. By exposing the tradeoffs between soft and hard RISC-V orchestration, RISC Bench demonstrates the importance of synchronization behavior, memory residency, and sustained throughput as first-class performance constraints.

Looking forward, RISC Bench is positioned as a proxy for future co-integrated architectures, including CPU–GPU packages, FPGA–GPU devices, FPGA–ASIC systems, and emerging FPGA–Quantum designs. The planned hybrid FPGA+accelerator testbed, benchmark suite extensions, and open leaderboard will expand RISC Bench’s relevance and foster community adoption.

In summary, RISC Bench offers a platform-independent, orchestration-aware descriptor that complements existing FLOP- and energy-based metrics. By grounding evaluation in sustained throughput rather than peak specifications, RISC Bench provides the FPGA and accelerator community with a practical tool to guide integration, optimize orchestration, and shape the next generation of heterogeneous computing systems.

ACKNOWLEDGMENTS

This work was supported in part by access to Altera Agilex FPGA development resources and the Tenstorrent Wormhole accelerator test environment. The authors also acknowledge the RISC-V community, whose open ecosystem provided the foundation for developing and validating RISC Bench.

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